

SURGICAL BASIN PORTFOLIO ELEMENT C

PRESENTATION AND JUSTIFICATION OF SOLUTION DESIGN REQUIREMENTS

Priority	Design Requirement	Why	Research
1 (High)	Ergonomic Comfort and Contour Fit	The basin needs to be able to support the shape of an arm or leg without causing pressure points of bruising. Comfort is the top concern for patients during the process of draining. A smooth, contoured support will reduce pain and improve patient safety.	<i>Medline Emesis Basin (U.S. Plastic Corp.)</i> and <i>Curved Surgical Bowl – JMU Hospital Supply</i> emphasize ergonomic curvature for comfort and spill prevention but still cause pressure buildup. Source Source
2 (High)	Stable Positioning During Use	During surgical fluid drainage, the basin should not shift or tilt. Stability prevents spills and contamination in sterile environments.	<i>Kangaroo Surgical Basin – BD Medical</i> and <i>Disposable Kidney Dish Review (PubMed)</i> both note that instability and slipping during procedures are common issues. Source Source
3 (High)	Sterility and Easy Sanitation	Preventing cross contamination is critical for safety. The basin must allow easy disinfection or disposal as well as hospital grade sterility.	<i>FDA Medical-Grade Basin Standards (2023)</i> show single-use polypropylene and smooth surfaces reduce pathogen risk. Our model uses a medical polymer or stainless steel with seamless edges for better cleaning. Additionally, hospitals and operating rooms use a sanitizing method called autoclaving to clean their tools and utensils. Ideally, the basin would be able to be sanitized through autoclaving. Source Source Source
4 (High)	Fluid Drainage Efficiency	The basin must allow fluid to drain all the way without and build up, which ensures surgeons can work quickly and efficiently.	Research on <i>Disposable Irrigation Basins – Cardinal Health</i> found that flat-bottom designs cause fluid stagnation. Incorporating an angled base or integrated drain channel will improve drainage performance. Source
5 (High)	Safety for Patient and Surgeon	The edges and materials cannot injure skin or cause allergic reactions to the surgeon or patient. Sharp edges can harm	<i>Medical Equipment Ergonomics Study (NIH, 2021)</i> found that smooth, rounded polymer edges lower injury risk. Our basin integrates rounded rims and

		tissues during movement.	skin-safe materials. Source
6 (Med)	Lightweight and Portable	For easy reposition during procedures, the basin must be light enough for the surgical staff.	<i>BD Kangaroo Disposable Basins</i> highlight that heavy stainless designs fatigue staff and slow workflow. We'll balance weight and stability using lightweight composite polymers. Source
7 (Med)	Cost-Effective Manufacturing	For hospitals to mass produce and replace frequently if disposable, the basin must be affordable.	<i>Cardinal Health Product Overview (2022)</i> showed polypropylene is cheaper than steel while maintaining durability. Our design uses injection-molded polymer for low cost. Source
8 (Med)	Spill-Resistant Rim Design	The basin's rim should minimize fluid overflow when draining from an extremity.	<i>Curved Kidney Basin Review (PubMed)</i> identified narrow rims as a spill hazard. A wider, angled lip or partial cover can reduce mess. Source
9 (Low)	Aesthetic/ Professional Appearance	The design (basin) should have a professional and modern appearance for use in sterile environments, however this is not a necessity.	<i>U.S. Plastic Corp. Design Guides</i> indicate that visual appeal increases staff satisfaction and perception of quality. Our model will have a clean, medical-grade finish. Source
10 (Low)	Recyclable or Sustainable Materials	Environmental responsibility is a growing concern in hospitals and their purchasing decisions, however, it is still one of the last things hospitals look for when making purchases.	<i>NIH Sustainable Medical Equipment Initiative (2023)</i> supports recyclable polypropylene. Our design can use recyclable plastics or biodegradable materials when feasible. Source Source